ADA University, SITE, Fall 2025, Discrete Structures

Midterm 1 (October 25, 2025). **SOLUTION SHEET**

27 points

Hint.

- Special sequences for the case of THREE input variables A, B, C (or x, y , z):

A (or x) = 00001111, B (or y) = 00110011, C (or z) = 01010101

- The truth table for the minterms (**using the special sequences for x, y, z**) are as follows:

Table 1

Truth tables for minterms using the special sequences for x, y, z

xyz=00000001	xyz′=00000010	xy′z=00000100	xy′z′=00001000
x′yz=00010000	x′yz′=00100000	x′y′z=01000000	x′y′z′=10000000

Tasks of Midterm I and Solutions.

Task 1 (4 pt). Design a three-input **minimal** AND-OR circuit L with the following truth table:

T =[x, y, z; L]=[00001111, 00110011, 01010101; **10011011**]

Hint.

Step 1. Find an appropriate Boolean expression as a sum of minterms (that is in CSoP form). (**1 pt**)
Step 2. Create a Karnaugh Map based on result of the Step 1. (**1pt**)
Step 3. Find a MSoP expression (**1 pt**)
Step 4. Design a three-input **minimal** AND-OR circuit L which corresponds to the Step 3 (**1 pt**)

Solution.

Step 1. (1 pt) Input sequences for the given circuit are special sequences. Represent the output sequence of bits generated by the given circuit, namely, **10011011** as a sum of minterms according to Table 1. We have **10011011**=x′y′z′+x′yz+xy′z′+xyz′+xyz

Step 2. (1 pt) Put checks in K-map according to the last expression.

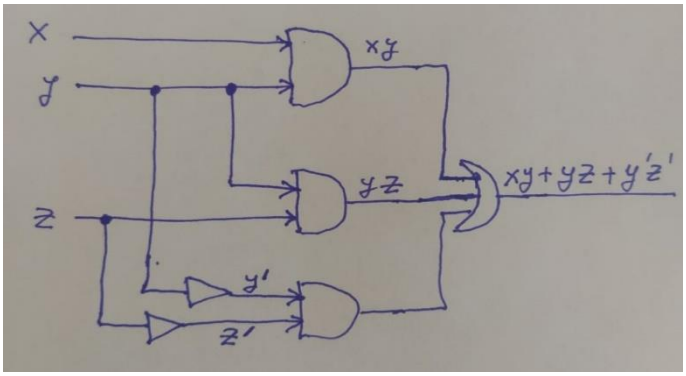
(A)	yz	yz′	y′z′	y′z
x	V	V	V	
x′	V		V	

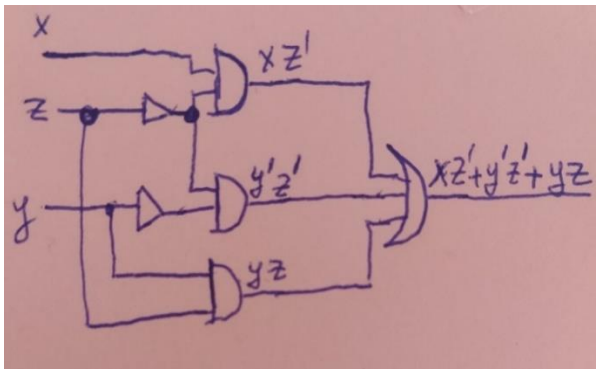
Step 3. (1 pt) MSoP= yz+y′z′+xy or MSoP=yz+ y′z′+ xz′

(It will be enough if the student suggests only one option).

Step 4. (1 pt) **Minimal** AND-OR circuit L which corresponds to is MSoP

(See next page)





MSoP= yz+y′z′+xy

MSoP= yz+ y′z′+ xz′

(It will be enough if the student suggests only one option).

Task 2 (5=3+1+1 pt).

Find the truth table T=T(E) for the Boolean expression

$$E=E(x, y, z)=(xy'+z')'(x+yz')+(x'z+y)(y'z)'+x'z+xy'z'$$

Hint.
Step 1. Find SoP form for E. (**3 pt**)
Step 2. Find CSoP form (**1pt**)
Step 3. Use Table 1 to find T(E) (T(E) will be the sequence of eight digits 0 or 1). (**1 pt**)

Solution.

Step 1. (3 pt). Find SoP form for E.

E=E(x, y, z)=(xy′+z′)′(x+yz′)+(x′z+y)(y′z)′+x′z+xy′z′ =

[(xy′)′z(x+yz′)]+[(x′z+y) (y+z′)]+x′z+xy′z′=

[(x′+y)z(x+yz′)]+[x′yz+x′zz′+yy+yz′]+x′z+xy′z′ =

[x′zx+x′zyz′+yzx+yzyz′]+[x′yz+y+yz′]+ x′z+xy′z′ =

[xyz]+[x′yz+y + yz′] + x′z+xy′z′ = xyz + y + x′z+xy′z′ = **y + x′z+xy′z′**

Step 2. (1 pt). Find CSoP form for E.

y + x′z+xy′z′ = (x+x′)y(z+z′) + x′(y+y′)z+xy′z′=

=[xyz + xyz′ + x′yz + x′yz′] + [x′yz + x′y′z] + xy′z′=

=xyz′+x′yz′+xyz+x′yz+x′y′z+xy′z′.

Step 3. (1 pt). Find T(E)

T(E)= xyz′ + x′yz′ + xyz + x′yz + x′y′z + xy′z′=0000 0010+0010 0000+0000 0001+

+0001 0000 + 0100 0000 + 00001000 =**0111 1011**

Answer: T(E)= **0111 1011**

Task 3 (4 pt=1+3)

Find a Boolean expression E=E(x, y, z) in **MSoP form** corresponding to the truth table T(E):

T(E)=11011001

Hint.

1. Find CSoP form for E by using Table 1. (**1 pt**)
2. Then use Karnaugh map method to reduce CSoP form to an MSoP form. It is enough to find at least one MSoP form if there are several such forms. (**3 pt**)

Solution.

Step 1. (1 pt). Find CSoP form for E by using Table 1.

CSoP(E)=x′y′z′+ x′y′z + x′yz + xy′z′+xyz

Step 2. (3 pt).

	yz	yz′	y′z′	y′z
x	V		V	
x′	V		V	V

It is easy to see that

MSoP= yz+y′z′+x′y′ OR MSoP= yz+ y′z′+ x′z

(It will be enough if the student suggests only one option).

Task 4 (8=4+4 pt)

1.Find a MSoP for the Boolean expression E given by Table (A). (*Write as an expression*). **(4pt)**
2.Find a MSoP for the Boolean expression E given by Table (B). (*Write as an expression*). **(4pt)**

(A)	**zt**	**zt′**	**z′t′**	**z′t**
xy			V	V
xy′	V	V	V	
x′y′		V		
x′y	V		V	V

(B)	**zt**	**zt′**	**z′t′**	**z′t**
xy	V		V	V
xy′	V	V	V	V
x′y′			V	
x′y	V		V	V

Solution.

MSoP=yz′+x′yt+xy′z+y′zt′+xz′t′ **MSoP=z′t′ + yt + xy′**
or
MSoP=yz′+x′yt+xy′z+y′zt′+xy′t′

(It will be enough if the student suggests only one option).

Task 5. (2 pt). Let S be a universal set and A and B are subsets of S. Prove De Morgan's law:

$(A\cup B)'=A'\cap B'$

Solution.

Let $x\in(A\cup B)' \Leftrightarrow x\in \mathbf{U}\text{-}(A\cup B) \Leftrightarrow x\notin(A\cup B) \Leftrightarrow x\notin A$ and $x\notin B \Leftrightarrow x\in A'$ and $x\in B' \Leftrightarrow x\in(A'\cap B')$

Task 6 (2 pt =1+1)

Subtask 6.1 (1 pt). Find the negation of the proposition below

$\exists x \exists y \forall z P(x, y, z)$

Subtask 6.2 (1 pt). Domain of all predicates P(x) below is the set of *all integers.* What are the truth values (True or False) of propositions below?

a) $\forall x(x>51)$ **(0.25 pt)** b) $\exists !x(x^5=-1)$ **(0.25 pt)**

c) $\exists !x(2x^2+3=1)$ **(0.25 pt)** d) $\exists x(2x=4x+2)$ **(0.25 pt)**

Solution.

Subtask 6.1 (1 pt).

$\neg(\exists x \exists y \forall z P(x, y, z))= \forall x\neg(\exists y \forall z P(x, y, z))= \forall x \forall y\neg(\forall z P(x, y, z))=$

$=\forall x \forall y \exists z(\neg P(x, y, z))$

Subtask 6.2 (1 pt).

a) False b) True c) False d) True

Task 7 (2 pt). It is known that the union of finite or countable number of countable sets is countable. Based on this result prove that the set A of all polynomials with rational coefficients is countable.

Proof. Let Q be the set of all rational numbers. A set P of all polynomials with rational coefficients is a disjoint union of the of subsets A_n, where A_n is the subset of all polynomials of degree equal n, that is,

$A_n=\{a_nx^n+a_{n-1}x^{n-1}+\ldots+a_1x+a_0 \mid a_i\in Q,\ i=0, 1, 2\ldots, n;\ \mathbf{a_n\neq 0}\}$, $n>0$; and $A_0=Q$ (that is A_0 is countable).

For each A_n, $n>0$, we have:

- under chosen $a_n, a_{n-1}, \ldots a_1$ the number of polynomials is countable (only a_0 is changes in Q);

Now we choose $a_n, a_{n-1}, \ldots, a_2$ and change a_1 in Q. For each a_1 number of choices for a_0 is countable. Hence, if we change a_1 and a_0 in Q then number of corresponding polynomials is the countable union of countable number polynomials and therefore is countable. By similar reasons we obtain that A_n is countable. Finally, since $P=\sum_{n=0}^{\infty} A_n$, so P is countable.

Good Luck!
Fuad bey